Shift-Left Testing

Key Takeaways:

- Shift testing activities earlier in the SDLC
- Automate as much as possible
- tighter feedback loops
- earlier defect identification & defect prevention
- shift left requires buy-in and collaboration among all team member: quality is a team responsibility
- Continuous Testing as part of the CI/CD process

Resources

- Your Shift left Initiatives are Failing - Here's How to Fix Them - Rich Jordan
  - John Ferguson Smart BDD
  - shift more focus Verification
  - Martin Fowler's Technical Debt Quadrants
  - Model Driven Development
- How BDD Can Help You Shift Left | TestLeft Webinar
  - Behavior Driven Development
- Four Types of Shift Left Testing
  - "*shift left testing*, which simply means beginning testing as early as practical in the lifecycle."
  - "testers can employ four fundamentally-different approaches to shift testing to the left"
  - "This blog post explains the importance of shift left testing and defines each of these four approaches using variants of the classic V model to illustrate them."
  - "Traditionally, people have decomposed *verification* into four different methods: *analysis*, *demonstration*, inspection, and *testing*."
    - Four methods of verification: 5.3 Product Verification
  - "On the other hand, as long as system and software engineers remember its limitations and view it merely as notional and only showing logical relationships between development activities, the V model provides a succinct way to illustrate our descriptions of approaches

to shift left testing. With this in mind, shift left essentially means moving testing to the left on the V.“

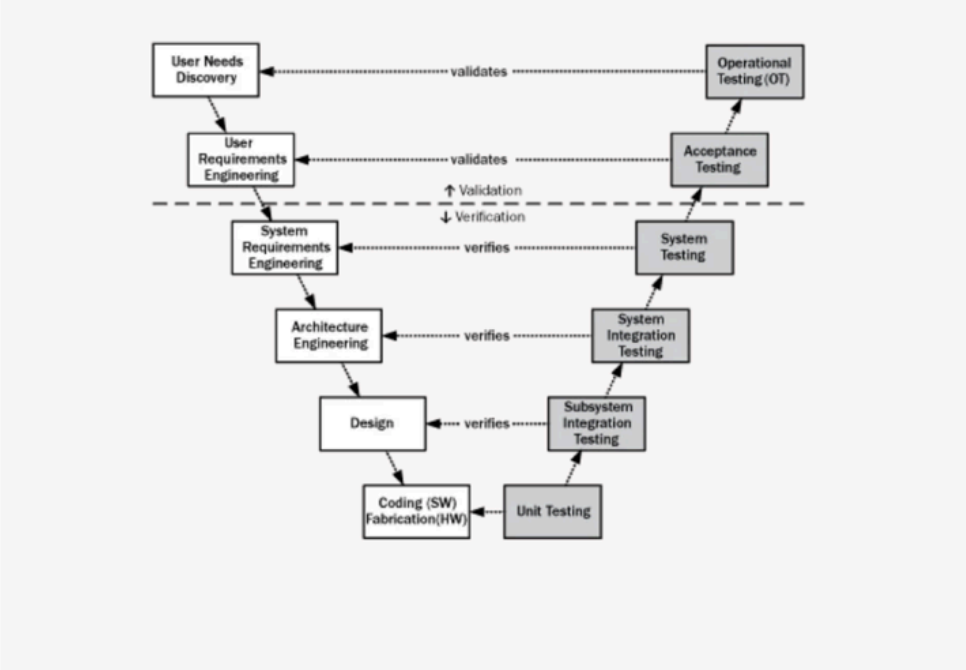
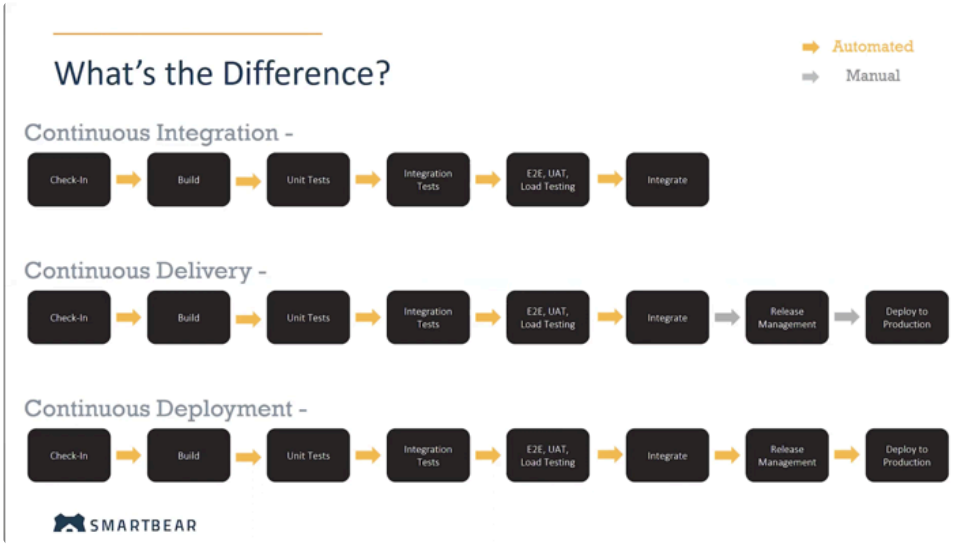


Figure 1: Traditional V Model

- What is Shift-left Testing? | IBM
  - “Shift-left testing is an approach in software development that emphasizes moving testing activities earlier in the development process.“
  - “Shift-left isn’t just about automation. Another way to test earlier and more often is to make sure that your QA specialists are involved in every step of your process, beginning with discovery and requirements gathering. Test engineers can do better when they have a greater understanding of the overall implementation, and their insights can help make the architecture more open and resilient.“
  - “If we think of TDD as “shift-left for monoliths,” then continuous testing is “shift-left for distributed architectures.”“
    - What Is Continuous Testing? | IBM
  - “Shift-left involves moving the testing activities closer to the beginning of the software development lifecycle, enabling faster feedback and reducing the time and effort required for bug fixing.“
- Shift Left Testing in Software Development
  - “Bringing development and testing together early is commonly referred to as ‘shifting left’.“
  - “Shift Left is a practice intended to find and prevent defects early in the software development process. The idea is to improve quality by moving tasks to the left as early in the lifecycle as possible.“

    - Reduced costs: “A defect that is removed after the product has gone into product will cost around 100 times more than one that is identified and removed during the requirements phase.“
    - What is Continuous Testing?
- What Is Shift Left Testing? A Guide to Improving Your QA
    - “The **“shift left” testing movement** is about pushing testing toward the early stages of software development. … The goal is to not find any critical bugs during the deployment phase that require code patching.“
    - Continuous Testing: “However, don’t think shift-left testing advocates that testing should be a discrete phase, only placed at the start of development, instead of at its end. As you probably know, in true software agile development, there shouldn’t be phases, but instead continuous activities occurring in short, iterative cycles.“
    - Behavior Driven Development
- Part 1 Your Complete Continuous Testing How To Guide



    - Behavior Driven Development
- A4Q Testing Summit 2025 - Synergizing TMMi and CI/CD: Continuous Test Process Improvement
- [Webcast] Enabling Continuous Testing in an agile paced DevOps environment(Part-3)
- What is Continuous Testing?
    - “Continuous testing is implemented into the development process so that QA happens every step of the way to ensure risk is constantly measured and mitigated.“
    - “Embedding testing into the software development process from beginning to end ensures that issues are found sooner and are more readily manageable“

    - “The pursuit of continuous testing is to eradicate bottlenecks completely by performing the rights tests at the right stages of each development cycle.“
    - “Continuous testing is a cultural shift from testing at the end to testing early, often, and at all stages of development with the utilization of automation wherever possible.“
    - Continuous testing is the process of incorporating automated feedback at different stages of the software development life cycle (SDLC) in support of better speed and efficiency when managing deployments.
- What Is Continuous Testing? | IBM
    - “Continuous testing works by using automated tools to load predefined quality assurance (QA) scripts at all stages of production.“

Note

I’m noticing connections/overlap between the following concepts:
Verification & Validation ⟷ Shift-left testing ⟷ SDLC

- Shifting testing earlier in the development lifecycle means including testers in the Verification process (“Are we building the right thing?“), whereas testers usually primarily operate within the Validation process (“Did we build the thing right?“)